

Antipresupposition: Turkish -sA Unified

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Intro to -sA (and -(y)sA) - The main goal

The so called conditional morpheme in Turkish comes in two varieties, according to previous literature (Göksel & Kerslake 2005, Soykan 2021): -(y)sA and -sA for indicative and subjunctive conditionals respectively. (Capitals denote harmonizing vowels).

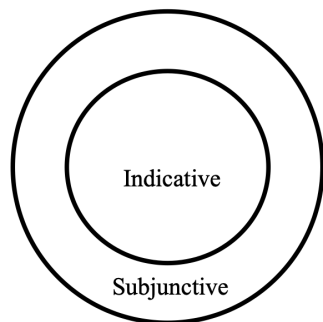
The main goal of this talk will be to find a way to unify the two conditional morphemes in Turkish and attribute the difference to the environment and morphemes that appear alongside these forms, namely, the non-past markers: the aorist/habitual -Ar, future -AcAK and progressive -Iyor)

X-marked and O-marked Conditionals

In many languages, the morphology of counterfactual/subjunctive (from now on referred as X-marking) conditionals are realized as a layer of past marking on top of the indicative conditional morphology (O-marking)

- (1) If John comes, Mary will go.
- (2) If John **came**, Mary **would** go. > X-marking, a layer of past

This also correlates with most analyses adding a special layer of meaning (X-mark) to reach the meanings, mimicking their morphology. The added meaning is usually analyzed as a form of domain widening (Stalnaker, 1968, 1975).



This domain widening effect has been analyzed either as modal (Iatridou 2000, Iatridou & von Stechow, 2023) or temporal (Ippolito, 2003, 2013). On the temporal side, the past marker directly correlates with an added past meaning in the derivation. Domain widening is obtained by going back in time and assuming a different situation where things have happened differently.

Indicative: [Indicative[Modal[if p], q]]

Subjunctive: [Past[Modal[if p], q]]

On modal side (Iatridou 2000, Iatridou & von Stechow, 2023 among many others), the main claim is that the past marker has a modal, non-temporal contribution that widens the domain of the evaluation worlds to include the worlds that are not compatible with the indicative.

Due to confusing terminology with aorist and the conditional morpheme(s), some remarks with regards to glossing in this paper is due:

- Aorist refers to the present tense marker in Turkish, the term aorist has been used for various grammatical markers cross-linguistically (usually perfective), due to this, I will gloss the aorist morpheme as **prs** (present), due to its similarities with what's called present tense in English.
- The conditional morpheme is analyzed as $-(y)sA$ and $-sA$ for O-marked and X-marked conditionals respectively, the morphological difference is due to the epenthetic phoneme y that appears before the O-mark counterpart in some environments where the preceding sound is a vowel. I will follow the literature on glossing and I will gloss the O-marked form as **YSA** and X-marked form as **SA**

Indicative Context (the speaker believes that p is possible)

Mary is waiting to decide whether to go to school. John texted her earlier saying he might come. If he does, she'll go with him. If not, she'll stay home. She's still unsure.

- (3) (Eğer) Can gel-**ir-se** Meryem okul-a gid-er
(If) John come-prs-YSA Mary school-dat go-aor
"If John comes, Mary will go to school."

- (4) # (Eğer) Can gel-**se** Meryem okul-a gid-er.
(if) John come-SA Mary school-dat go-aor
"If John came, Mary would go to school."

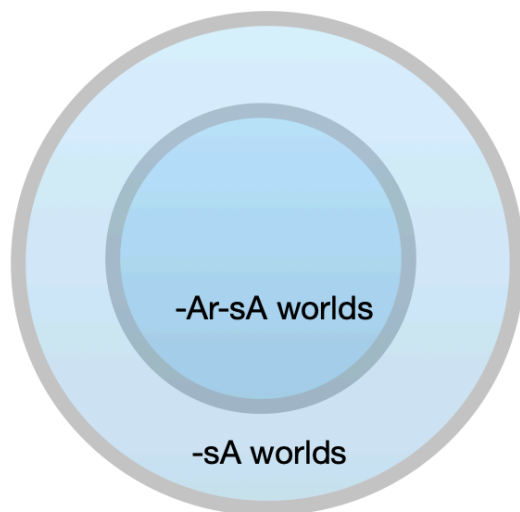
Note: The optional *if*-particle *eğer* is much degraded in X-marked conditionals. The reason is not clear to me, but subordinating particles can be specified for specific type of conditionals in other languages as well, one example is Czech with *jestli* for O-marked, *kdyby* for X-marked conditionals.

Subjunctive Context (the speaker believes that p is less likely than $\neg p$)

John is usually too lazy to come to school these days, and today is no exception—he’s probably at home sleeping. Mary dislikes going to school unless John is there, but since he’s unlikely to come, she’s staying home too. The speaker says:

- (3) # (Eğer) Can gel-**ir-se** Meryem okul-a gid-er
 (If) John come-prs-YSA Mary school-dat go-aor
 “If John comes, Mary will go to school.”

- (4) Can gel-**se** Meryem okul-a gid-er.
 John come-SA Mary school-dat go-aor
 “If John came, Mary would go to school.”



Turkish also doesn't have the morphological ambiguity wrt. past oriented O-marked conditionals and X-marked counterfactuals/subjunctives:

Counterfactual Context (the speaker knows p is not possible, and reasons what could have happened if p were the case):

John flew to Paris last week and was not home yesterday, and Mary broke both her legs so she cannot go to school by herself, the speaker says:

- (5) # (dün) Can ev-e gel-**di-yse** Meryem okul-a
 (yday) John home-dat come-PST-YSA Mary school-dat
 git-**miş**-tir
 go-**EVID.PST-AOR**
 "If John came home yesterday, Mary must have gone to school"

- (6) (dün) Can ev-e gel-**se-ydi** Meryem okul-a
 (yesterday) John home-dat come-SA-PST Mary school-dat
 gid-er-di
 go-prs-pst
 "If John had come home yesterday, Mary would have gone to the school."

Past Indicative Context (the speaker doesn't know if p happened in the past, but deducts that q must have happened if p did):

We don't know if Mary went to school yesterday, but we know that Mary goes to school when John comes home. While speaking about Mary's attendance to school yesterday, the speaker says:

- (5) (dün) Can ev-e gel-**di-yse** Meryem okul-a
 (yday) John home-dat come-pst-YSA Mary school-dat
 git-**miş**-tir
 go-**EVID.PST-AOR**
 "If John came home yesterday, Mary must have gone to school"

- (6) # (dün) Can ev-e gel-se-ydi Meryem okul-a
 (yesterday) John home-dat come-SA-pst Mary school-dat
 gid-er-di
 go-prs-pst

“If John had come home yesterday, Mary would have gone to the school.”

Note: inferential/evidential past seems to be obligatory in past indicatives/epistemic conditionals since the speaker cannot directly know about Mary’s actions.

And the subjunctive form doesn’t accept past adverbials, as expected.

- (4’) (*dün) Can gel-se Meryem okul-a gid-er.
 (*yesterday) John come-SA Mary school-dat go-prs
 “If John came, Mary would go to school.”

Lastly, -sA form antecedents are also used in desiderative settings, marking the antecedent clause as the complement of a desiderative verb.

- (7) Can gel-se çok sevin-ir-im
 John come-SA very happy.be-prs-1sg
 ‘I’d be very happy if John came.’

Or by themselves with a covert consequent, optionally marked with the desiderative particle *keşke* (borrowed from Persian) which shows complementary distribution with if-particle *eğer* wrt. The O-marked and X-marked conditional particles.

- (8) Keşke Can gel-se.
 if.only John come-SA
 ‘If only John came.’
 (9) * Keşke Can gel-ir-se.
 if.only John come-prs-YSA
 ‘If only John came.’

Desideratives has been noted to have morphological connection to counterfactuals in many languages (Iatridou, 2000). Turkish is an example of such a language.

Previous Analyses of -sA (and -(y)sA)

Göksel & Kerslake 2005:

They analyze -sA and -(y)sA as two separate morphemes, corresponding to subjunctive and indicative respectively.

Turkish has two layers of morphemes that can contribute temporal/aspectual meanings, where the elements can be used to build the meaning like lego blocks.

Verb	Slot 1	copula	Slot 2
Git	-AcAK	(y)	-DI
Go	Future		Past
	-mİş		-mİş
	Perfective		Evidential
	-Iyor		-sA
	Progressive		Conditional (O-mark)
	-Ar		
	Present/Aorist		
	-sA		
	Conditional (X-mark)		
	-dİ		
	Past		

The first layer is called the aspectual layer, where the combination of verb and the aspectual morpheme results in a non-verbal element, which requires the insertion of copula to be verbalized again.

Only a few combinations are blocked:

*sa+sa

*miş+miş (depends on the dialect, not available on standard)

The past-oriented indicative conditional is the only form where the past marker -dİ appears in the lower layer.

The copular suffixes (the higher layer) can be combined with nouns as well:

- (10) Bu bir elma-ysa onu ye.
 this an apple-YSA this.acc eat.imp
 ‘If this is an apple, eat it.’

- (11) Bu bir elma-ymış / elma-ydı.
 this an apple-evid apple-pst
 ‘This was (allegedly) an apple.’

Lastly, two form of -sA behave differently wrt. nouns: (compare with 10, which is O-marked)

- (12) Bu bir elma ol-sa, onu ye-r-di-m
 this an apple be(come)-SA this.acc eat-prs-pst-1sg
 ‘If this was an apple, I would eat it.’

Göksel & Kerslake (2005) propose that copula and the **aorist** don’t add any meaning, only there to support -(y)sA structurally, similar to how the auxiliary is inserted to support the CF form in nouns.

Copular suffixes can optionally be written separately from the verbal complex:

- (13) Can gel-ecek ise Meryem gel-me-yecek.
 John come-fut cop.YSA Mary come-neg-fut
 ‘If John is going to come, Mary isn’t going to come.’

These copular cases are one evidence for Göksel & Kerslake (2005)’s analysis. Their analysis is based on this copular element, which is phonologically weakened to ‘y’ when the copular suffix is combined with the verb according to them.

However, the appearance of the so called copula even with a low suffix like the future marker in (example above) casts a doubt on whether this ‘y’ morpheme is actually an element or just a phonological insertion to break up hiatus.

> This is a doubly ambiguous approach since aorist has meaning in all other cases:

- (14) Yürü-r.
walk-prs
“He (habitually) walks.”

- (15) Yürü-r-dü.
walk-prs-pst
“He used to walk.” > habitual+past

Soykan 2021 and 2023:

Soykan (2021) proposes an ambiguity approach based on presuppositions. She claims that -sA and -(y)sA are different morphemes with different presuppositions.

- (16) **Presuppositions of Present Subjunctive Conditionals:** p-sA, q

- a. p is more likely to be false than true for s
- b. p is epistemically possible for s
- c. q is epistemically possible for s

- (17) **Presuppositions of Present Indicative Conditionals:** p-ysA, q

- a. p is epistemically possible for s
- b. q is epistemically possible for s

As for the aorist, she analyzes that its distribution in antecedents is restricted by the consequent, to the cases where the conditional is future oriented and indicative.

> She claims a meaning for the aorist, which fares better than Göksel&Kerslake (2005)

> While there is nothing inherently wrong with this analysis, it's still an ambiguity approach.

Solution to the Dichotomy: Antipresupposition

Recent literature on indicative conditionals analyze them as being presuppositionally heavier than subjunctives (Leahy 2011, **von Fintel 1999**). Specifically on Leahy (2011)'s analysis, the presupposition is **epistemic possibility for the speaker**.

Indicative:	$\Diamond_{\text{epis}_s} p$	(p is variable over propositions)
Subjunctive:	$\emptyset$	

Maximize Presupposition! (Definition below) dictates that the presuppositionally heavier alternative must be used if there are two forms competing with regards to their presuppositional

(18) Maximize Presupposition!

If a sentence S is a presuppositional alternative of a sentence S0 (i.e., $S \text{ [Alt}(S0))$ and the context C is such that:

- (i) the presuppositions of S and S0 are satisfied within C;
 - (ii) S and S0 have the same assertive component relative to C;
 - (iii) S carries a stronger presupposition than S0,
- then S should be preferred to S0.

Similar to scalar implicature that obtains with competing assertive components, competing presuppositional components also give rise to an implicature, which can be best described as scalar inference for presuppositions, called Antipresupposition (19)

(19) Antipresupposition:

Given Maximize Presupposition!, if the sentence S0 with weaker presuppositions are uttered by the speaker instead of sentence S with stronger presuppositions, then it's possible to say that the speaker sees the presuppositions of S as not met.

Assuming the speaker authority, competence and reliability, the following implicature obtains in non-indicative conditionals:

- | | | |
|------|---|------------------------------|
| i. | $\neg B_s[\Diamond_{\text{epis}_s} p]$ OR $\neg B_s[\text{Auth}[\Diamond_{\text{epis}_s} p]]$ | |
| ii. | $\neg B_s[\Diamond_{\text{epis}_s} p]$ | (from (i) and Authority) |
| iii. | $B_s[\sim \Diamond_{\text{epis}_s} p]$ | (from (ii) and Competence) |
| iv. | $\neg \Diamond_{\text{epis}_s} p$ | (from (iii) and Reliability) |
- Leahy (2011)

For this account, assuming that the aorist morpheme (and all the other non-past morphemes) in Turkish carries the epistemic possibility presupposition. This change from Soykan (2021)’s analysis on -(y)sA and -sA’s presuppositions unifies their meaning.

I am agnostic towards the exact meaning of -sA (unified) as of yet, this account does not depend on a specific account of conditional marker itself in Turkish.

Open questions:

- Pragmatic competition is said to arise only when the structures are of equal or lesser complexity. (Katzir, 2007) But in case of Turkish conditionals, the indicative form seems to have more morphology.

Assuming a phonologically null subjunctive morpheme below -sA in subjunctive cases bring the structures to equal structural complexity.

- (20) Can gel-ir i-se, gid-er-iz
 John come-prs be-YSA go-aor-1pl
 ‘We’ll go if John comes.’ (Indicative past)

- (21) Can gel-Ø-se, gid-er-iz
 John come-sbj-SA go-prs-1pl
 ‘We would go if John came.’

The same subjunctive morpheme could be assumed in imperative cases

- (22) Bura-ya gel-Ø
 here-dat come-subj/imp
 ‘Come here.’

● Counterfactual Cases

If the future-less-vivid (FLV) has no presuppositions and by antipresupposition it receives the epistemically impossible meaning, then what about CF?

Leahy (2011) doesn’t distinguish between CF and subjunctive cases, clumping them both together under non-indicative. However, adopting such an approach without distinguishing between the types of subjunctive can be problematic.

Adopting Ippolito (2013)’s presuppositional approach to account for the CF cases could work. This will be worked out in the future.

Indicative/O-marked: $\Diamond_{\text{epis}_s} p$

Subjunctive/X-marked: $\emptyset$

CF/Double X-marked: $\neg p \in \text{CG}$

Presuppositions in CF cases:

Only in cases where $\neg p$ is in the CG/speaker’s can the speaker utter the CF form:

Context: Tomorrow is John’s birthday, but I confused the dates and called him today and that made him really upset.

- (25) #Can-1 yarın ara-**sa-m** daha iyi ol-ur-du
 John-acc tomorrow call-sa-1sg more good be-prs-pst
 ‘If I called John tomorrow, it would have been better.’

- (26) Can-1 yarın ara-**sa-ydı-m** daha iyi ol-ur-du
 John-acc tomorrow call-sa-pst-1sg more good be-prs-pst
 ‘If I had called John tomorrow, it would have been better.’

Here, even though the event in the antecedent is about the future, one layer of X-marking cannot be used because the calling event has already occurred. So it's not an accessible world from the utterance time.

This creates a problem for Leahy (2011)'s approach, since the implicature already derives $\neg\Diamond_{\text{epis}_s} p$. Ippolito (2008) assumes that Counterfactual/Double X-marking is used in cases where the presuppositions of the antecedent proposition is not met:

- In subjunctive cases, even if the eventuality described in the antecedent is counterfactual, presuppositions in the antecedent (if any) cannot be.
- If the antecedent has no presuppositions, subjunctive is always felicitous.
- When one of the presuppositions in the antecedent is not met, CF is required.

Conclusion:

- *-sA* by itself includes the worlds that are also not in speakers doxastic alternatives. Adding *-Ar* limits the worlds to the ones that are epistemically possible according to the speaker.
- Turkish is a language where the X-marking works inversely (O-mark has more overt marking than X-mark).
- An antipresuppositional analysis for indicative conditionals in Turkish can account for the two conditional morphemes in Turkish without postulating an ambiguity approach.
- The unified antipresupposition analysis supports the idea that other high attaching suffixes (*-DI*, *-mİş*) also have a single meaning, further specified by the surrounding context and morphemes rather than analyzing each attachment site as different lexical entries.

	O vs X Explanation	Status of Cond Marker
Göksel & Kerslake	Structural difference	Ambiguity wrt. <i>-sA</i>
Soykan	Presuppositional Difference on <i>-sA</i>	Ambiguity wrt. <i>-sA</i>
Antipresupposition	Presuppositional Difference on Aorist	<i>-sA</i> Unified

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